



Mathe**magical**

Amaze your friends *and* family with *these*



Two wrongs make a right

When nobody's looking, take a sneaky peek at the top card in a pack (let's say it's the 10 of hearts). Announce that you will memorize the entire pack by flicking through them once. Give them a quick flick, then hand them to a friend. Ask your friend to *think of a number* from 1 to 10 and deal out that many cards, face down in a pile. Say the next card is the 10 of hearts and ask them to turn it over. It isn't, so pretend to be disappointed.

Tell them to put it back and place the small pile back on top. Ask for another number between 10 and 20, then try again, pretending to be disappointed a second time. Finally, ask your friend to subtract the first number from the second, and try one last time. Now it works!



The amazing magic calculator

Give a friend a calculator and ask them to punch in any 3-figure number twice to make a 6-figure number. Tell your friend that the chance of 7 dividing into a random number without a remainder is 1 in 7.

Ask them to try it. Any remainder? No.

That was lucky! Tell them to try dividing the number on screen by 11. The chance of this working is 1 in 11. Any remainder? No – amazing! Now try dividing by 13. Any remainder? No – *astonishing*! To finish off, ask what's left. It's the original 3-figure number!!! But why?



The mind-boggling 1089 trick

First do some preparation. Open a book on page 10, count down 8 lines and along 9 words. Write the 9th word on a slip of paper, seal it in an envelope, and place it on a table under the book. Now for the trick. Ask a friend to think of a 3-digit number and write it down. Any number will do as long

as the first and last digits differ by two or more. Tell your friend to reverse the number and subtract the smaller one from the bigger one. For instance, $863 - 368 = 495$. Then reverse the digits in the answer and add the two numbers: $495 + 594 = 1089$. Now tell your friend to use the first two digits in the answer as the page of the book. They should use the 3rd digit to find the line, and the last digit to find the word. Tell them to read the word out loud. Finally, ask your friend to open the envelope. This trick works because the answer is *always* 1089!



- Give a friend a calculator and tell them to key in the number of the month in which they were born
- Multiply by 4
- Add 13
- Multiply by 25
- Subtract 200
- Add the day of the month they were born
- Multiply by 2